

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 7: Ratio and Proportion

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Properties of proportion

1. Use the properties of proportion to solve $\frac{6x^2 + 3x - 5}{3x - 5} = \frac{9x^2 + 2x + 5}{2x + 5}$, where $x \neq 0$. **ICSE 2025, 3 marks**

SOLUTION

Apply dividendo: if $\frac{a}{b} = \frac{c}{d}$ then $\frac{a-b}{b} = \frac{c-d}{d}$.

$$\frac{6x^2 + 3x - 5 - (3x - 5)}{3x - 5} = \frac{9x^2 + 2x + 5 - (2x + 5)}{2x + 5}, \text{ so } \frac{6x^2}{3x - 5} = \frac{9x^2}{2x + 5}$$

Since $x \neq 0$, divide both sides by $3x^2$: $\frac{2}{3x - 5} = \frac{3}{2x + 5}$

$$4x + 10 = 9x - 15, \text{ so } 5x = 25.$$

Answer: $x = 5$

2. Given $\frac{(a+b)^3}{(a-b)^3} = \frac{64}{27}$, find (a) $\frac{a+b}{a-b}$ (b) the ratio $a : b$, using the properties of proportion.

ICSE 2024, 4 marks

SOLUTION

(a) Take the cube root of both sides: $\frac{a+b}{a-b} = \frac{4}{3}$

(b) Apply componendo and dividendo: $\frac{(a+b) + (a-b)}{(a+b) - (a-b)} = \frac{4+3}{4-3}$

$$\frac{2a}{2b} = \frac{7}{1}$$

Answer: (a) $\frac{4}{3}$ (b) $a : b = 7 : 1$

3. Use componendo and dividendo to solve $\frac{\sqrt{2x+2} + \sqrt{2x-1}}{\sqrt{2x+2} - \sqrt{2x-1}} = 3$. **ICSE 2023, 3 marks**

SOLUTION

Write 3 as $\frac{3}{1}$ and apply componendo and dividendo: $\frac{2\sqrt{2x+2}}{2\sqrt{2x-1}} = \frac{3+1}{3-1} = 2$

Square both sides: $\frac{2x+2}{2x-1} = 4$, so $2x+2 = 8x-4$

$$6x = 6$$

Answer: $x = 1$

4. Find $x : y$, given $\frac{x^2 + 2x}{2x + 4} = \frac{y^2 + 3y}{3y + 9}$. **ICSE 2020, 3 marks**

SOLUTION

Factorise: $\frac{x(x + 2)}{2(x + 2)} = \frac{y(y + 3)}{3(y + 3)}$

$\frac{x}{2} = \frac{y}{3}$. By alternendo, $\frac{x}{y} = \frac{2}{3}$.

Answer: $x : y = 2 : 3$

5. If $x = \frac{\sqrt{2a+1} + \sqrt{2a-1}}{\sqrt{2a+1} - \sqrt{2a-1}}$, prove that $x^2 - 4ax + 1 = 0$. **ICSE 2020, 4 marks**

SOLUTION

Write x as $\frac{x+1}{x-1}$ and apply componendo and dividendo: $\frac{x+1}{x-1} = \frac{2\sqrt{2a+1}}{2\sqrt{2a-1}}$

Square both sides: $\frac{x^2 + 2x + 1}{x^2 - 2x + 1} = \frac{2a + 1}{2a - 1}$

Apply componendo and dividendo again: $\frac{2x^2 + 2}{4x} = \frac{4a}{2}$, so $\frac{x^2 + 1}{2x} = 2a$

$x^2 + 1 = 4ax$, that is $x^2 - 4ax + 1 = 0$. **Proved.**

6. Use the properties of proportion to solve $\frac{\sqrt{5x} + \sqrt{2x-6}}{\sqrt{5x} - \sqrt{2x-6}} = 4$. **ICSE 2019, 3 marks**

SOLUTION

Apply componendo and dividendo: $\frac{2\sqrt{5x}}{2\sqrt{2x-6}} = \frac{4+1}{4-1} = \frac{5}{3}$

Square both sides: $\frac{5x}{2x-6} = \frac{25}{9}$, so $45x = 50x - 150$

Answer: $x = 30$

7. Use the properties of proportion to solve $\frac{2x + \sqrt{4x^2 - 1}}{2x - \sqrt{4x^2 - 1}} = 4$, given that x is positive. **ICSE 2018, 3 marks**

SOLUTION

Apply componendo and dividendo: $\frac{4x}{2\sqrt{4x^2 - 1}} = \frac{4+1}{4-1}$, so $\frac{2x}{\sqrt{4x^2 - 1}} = \frac{5}{3}$

Square both sides: $\frac{4x^2}{4x^2 - 1} = \frac{25}{9}$, so $36x^2 = 100x^2 - 25$

$64x^2 = 25$, and x is positive.

Answer: $x = \frac{5}{8}$

8. Given $\frac{7m+2n}{7m-2n} = \frac{5}{3}$, use the properties of proportion to find (a) $m : n$ (b) $\frac{m^2+n^2}{m^2-n^2}$. **ICSE 2017, 3 marks**

SOLUTION

(a) Apply componendo and dividendo: $\frac{14m}{4n} = \frac{5+3}{5-3} = 4$, so $\frac{m}{n} = \frac{8}{7}$.

(b) $\frac{m^2}{n^2} = \frac{64}{49}$. Apply componendo and dividendo: $\frac{m^2+n^2}{m^2-n^2} = \frac{64+49}{64-49}$

Answer: (a) $8 : 7$ (b) $\frac{113}{15}$

2 Proportion

9. A train moves at a uniform speed. It covers 60 m in 2 s, 90 m in x s and y m in 5 s. Find x and y . (a) $x = 4, y = 150$ (b) $x = 3, y = 100$ (c) $x = 4, y = 100$ (d) $x = 3, y = 150$ **ICSE 2024, 1 mark**

SOLUTION

Speed = $\frac{60}{2} = 30$ m/s. At uniform speed, distance is proportional to time.

$x = \frac{90}{30} = 3$ and $y = 30 \times 5 = 150$

Answer: (d)

10. What number must be added to each of 4, 6, 8 and 11 so that the four numbers are in proportion?

ICSE 2023, 3 marks

SOLUTION

Let the number be x . Then $\frac{4+x}{6+x} = \frac{8+x}{11+x}$.

$(4+x)(11+x) = (6+x)(8+x)$, so $44 + 15x + x^2 = 48 + 14x + x^2$

$x = 4$. Check: 8, 10, 12, 15 gives $8 \times 15 = 10 \times 12 = 120$.

Answer: 4

11. The numbers $K+3, K+2, 3K-7$ and $2K-3$ are in proportion. Find K . **ICSE 2019, 3 marks**

SOLUTION

$\frac{K+3}{K+2} = \frac{3K-7}{2K-3}$, so $(K+3)(2K-3) = (K+2)(3K-7)$

$2K^2 + 3K - 9 = 3K^2 - K - 14$, so $K^2 - 4K - 5 = 0$

$(K-5)(K+1) = 0$

Answer: $K = 5$ or $K = -1$

3 Continued proportion

12. Find the mean proportional between 4 and 9. (a) 4 (b) 6 (c) 9 (d) 36 **ICSE 2023, 1 mark**

SOLUTION

$$\text{Mean proportional} = \sqrt{4 \times 9} = \sqrt{36} = 6$$

Answer: (b)

13. b is the mean proportional between a and c . Show that $\frac{a^4 + a^2b^2 + b^4}{b^4 + b^2c^2 + c^4} = \frac{a^2}{c^2}$. **ICSE 2017, 3 marks**

SOLUTION

b is the mean proportional, so $b^2 = ac$.

$$\text{Numerator} = a^4 + a^2(ac) + (ac)^2 = a^2(a^2 + ac + c^2)$$

$$\text{Denominator} = (ac)^2 + (ac)c^2 + c^4 = c^2(a^2 + ac + c^2)$$

$$\text{LHS} = \frac{a^2(a^2 + ac + c^2)}{c^2(a^2 + ac + c^2)} = \frac{a^2}{c^2} = \text{RHS. Proved.}$$

These questions are based on questions asked in past board examinations. They have been reworded by MathSetu, the figures are redrawn, and the solutions are MathSetu's own. MathSetu is an independent study resource and is not affiliated with or endorsed by CBSE, CISCE, NCERT or any publisher.